Chapter 8

Local comparison of functions

In this chapter we shall gather some tools that will allow us to study the *asymptotic* behavior of functions. The asymptotic behavior of a function $f(x)$ can refer either to its behavior as $x \to \pm\infty$, or as $x \to x_0 \in \mathbb{R}$, where $f(x)$ might tend to 0 or to $\pm\infty$.

8.1 Landau symbols

8.1.1 Definitions

Let $f, g : \mathbb{R} \to \mathbb{R}$ and let $x_0 \in \{-\infty\} \cup \mathbb{R} \cup \{+\infty\}$. We introduce the **Landau symbols**:

Big $\mathcal{O}$

If there exist a constant $C > 0$ and a deleted neighborhood of x_0 on which

$$|f(x)| \leq C|g(x)|,$$

then we say that f **is controlled by** g as $x \to x_0$, and we write

$$f = \mathcal{O}(g), \quad x \to x_0.$$

We often say that f **is big $\mathcal{O}$ of** g as $x \to x_0$. If $g(x) \neq 0$ on a deleted neighborhood of x_0, this simply means that $\frac{f}{g}$ is bounded on a deleted neighborhood of x_0.

For the remaining symbols, we assume that $g(x) \neq 0$ on a deleted neighborhood of x_0, and that the limit

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \ell$$

exists (can be finite or infinite).

Same order

If ℓ is finite and $\ell \neq 0$, then we say that f **has the same order of magnitude as** g as $x \to x_0$, and we write

$$f \asymp g, \quad x \to x_0.$$

Asymptotically equivalent

If $\ell = 1$, then we say that f **is equivalent to** g as $x \to x_0$, and we write

$$f \sim g, \quad x \to x_0.$$

Little o

If $\ell = 0$, then we say that f **is negligible with respect to** g as $x \to x_0$, and we write

$$f = o(g), \quad x \to x_0.$$

We often say that f **is little** o **of** g as $x \to x_0$.

$\ell = \pm\infty$

If $\ell = \pm\infty$, we need to look at $\frac{g}{f}$ instead!

Example 8.1:

1. Since $\lim_{x\to 0} \frac{\sin x}{x} = 1$ we have

$$\sin x \sim x, \quad x \to 0.$$

2. Since $\lim_{x\to+\infty} \frac{\sin x}{x} = 0$ we have

$$\sin x = o(x), \quad x \to +\infty.$$

3. Since $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$ we have

$$1 - \cos x \asymp x^2, \quad x \to 0.$$

4. Since $|\sin x| \leq 1$ for all $x \in \mathbb{R}$ we have

$$\sin x = \mathcal{O}(1), \quad x \to +\infty,$$

even though $\lim_{x\to+\infty} \sin x$ does not exist.

8.1.2 Basic properties

1. Observe that f and g having the same order, or being asymptotically equivalent, or being little o, are all subcases of being Big $\mathcal{O}$, that is:

$$\begin{aligned} f \asymp g \quad &\Rightarrow \quad f = \mathcal{O}(g) \\ f \sim g \quad &\Rightarrow \quad f = \mathcal{O}(g) \\ f = o(g) \quad &\Rightarrow \quad f = \mathcal{O}(g) \end{aligned}$$

   all as $x \to x_0$. Indeed, if $\lim_{x\to x_0} \frac{f(x)}{g(x)} = \ell \in \mathbb{R}$, then (taking $\varepsilon = 1$ in the definition of limit) $\left|\frac{f(x)}{g(x)}\right| \leq |\ell| + 1$ on a deleted neighborhood of x_0, so $f = \mathcal{O}(g)$. Another implication is:

$$f \sim g \quad \Rightarrow \quad f \asymp g$$

   as $x \to x_0$. Conversely:

$$f \asymp g \quad \Rightarrow \quad f \sim \ell g$$

   as $x \to x_0$.

2. $$f \sim g \quad \Leftrightarrow \quad f = g + o(g)$$

   as $x \to x_0$.

3. $$\begin{aligned} f = \mathcal{O}(g) \quad &\Leftrightarrow \quad f = \mathcal{O}(\lambda g),\ \forall \lambda \neq 0 \\ f = o(g) \quad &\Leftrightarrow \quad f = o(\lambda g),\ \forall \lambda \neq 0 \end{aligned}$$

   all as $x \to x_0$.

4. $$\begin{aligned} f = \mathcal{O}(1),\ x \to x_0 \quad &\Leftrightarrow \quad f \text{ is bounded on a deleted neighborhood of } x_0 \\ f = o(1),\ x \to x_0 \quad &\Leftrightarrow \quad f(x) \xrightarrow{x\to x_0} 0 \end{aligned}$$

   In particular, if $f(x) \to \ell \in \mathbb{R}$ as $x \to x_0$ (for instance if $f = o(1)$), then $f = \mathcal{O}(1)$. The converse is false: $\sin x = \mathcal{O}(1)$ as $x \to +\infty$, but $\sin x$ has no limit.

5. $$f \text{ is continuous at } x_0 \quad \Leftrightarrow \quad f(x) = f(x_0) + o(1),\ \text{as } x \to x_0.$$

8.1.3 Comparing powers

One of the simplest functions is the monomial power of x: x^n. We therefore want to be able to compare them near where they vanish ($x_0 = 0$) and near where they tend to infinity ($x_0 = \pm\infty$).

Near 0

Near $x_0 = 0$ we have:

$$x^n = o(x^m),\ x \to 0, \qquad \Leftrightarrow \qquad n > m,$$

since $x^{n-m} \to 0$ as $x \to 0$ when $n > m$. This implies that **near 0, bigger powers are negligible**.

Near $\pm\infty$

Near $x_0 = \pm\infty$ we have:

$$x^n = o(x^m),\ x \to \pm\infty, \qquad \Leftrightarrow \qquad n < m,$$

since $x^{n-m} = \frac{1}{x^{m-n}} \to 0$ as $x \to \pm\infty$ when $n < m$. This implies that **near $\pm\infty$, smaller powers are negligible**.

8.1.4 Algebra of little o

When we compute with asymptotic expressions (as we will do extensively with Taylor expansions), we need rules for manipulating the symbol o.

Notation

In a formula, $o(g)$ denotes *some* function which is $o(g)$ as $x \to x_0$ (we do not care which one). Therefore an identity such as $o(g) + o(g) = o(g)$ means: *the sum of any two functions that are $o(g)$ is again $o(g)$*. Such identities should only be read from left to right.

Warning!

We have $o(g) - o(g) = o(g)$, *not* 0: the two functions denoted by $o(g)$ need not be the same. For instance, $x^2 = o(x)$ and $2x^2 = o(x)$ as $x \to 0$, but $x^2 - 2x^2 = -x^2 \neq 0$ (it is, however, $o(x)$).

Proposition 8.1: Let $g : \mathbb{R} \to \mathbb{R}$ with $g(x) \neq 0$ on a deleted neighborhood of x_0, and let $c \in \mathbb{R}$. Then, as $x \to x_0$:

1. $o(g) \pm o(g) = o(g)$;
2. $c\,o(g) = o(g)$;
3. $h\,o(g) = o(hg)$, if $h(x) \neq 0$ on a deleted neighborhood of x_0;
4. $o(g)\,o(h) = o(gh)$, if $h(x) \neq 0$ on a deleted neighborhood of x_0;
5. $h\,o(g) = o(g)$, if h is bounded on a deleted neighborhood of x_0 (that is, $h = \mathcal{O}(1)$).

Proof. Let $f_1 = o(g)$ and $f_2 = o(g)$, that is, $\frac{f_1}{g} \to 0$ and $\frac{f_2}{g} \to 0$ as $x \to x_0$.

1. $\frac{f_1 \pm f_2}{g} = \frac{f_1}{g} \pm \frac{f_2}{g} \to 0 \pm 0 = 0$.
2. $\frac{cf_1}{g} = c\,\frac{f_1}{g} \to c \cdot 0 = 0$.
3. $\frac{hf_1}{hg} = \frac{f_1}{g} \to 0$.
4. If $f_3 = o(h)$, then $\frac{f_1 f_3}{gh} = \frac{f_1}{g}\,\frac{f_3}{h} \to 0 \cdot 0 = 0$.
5. If $|h| \le M$ on a deleted neighborhood of x_0, then there $0 \le \left|\frac{hf_1}{g}\right| \le M\left|\frac{f_1}{g}\right|$, and the right-hand side tends to 0. By the Squeeze Theorem, $\frac{hf_1}{g} \to 0$.

□

In practice, we mostly use these rules with monomials, as $x \to 0$:

Corollary 8.2: Let $n, m, k \in \mathbb{N}$ and let p be a polynomial with $p(0) \neq 0$. Then, as $x \to 0$:

$$\begin{aligned} o(x^n) + o(x^m) &= o(x^{\min\{n,m\}}) \\ x^k\, o(x^n) &= o(x^{n+k}) \\ o(x^n)\, o(x^m) &= o(x^{n+m}) \\ p(x)\, o(x^n) &= o(x^n) \end{aligned}$$

Proof. Suppose $n \le m$ (the other case is symmetric). Then every $f = o(x^m)$ is also $o(x^n)$, since $\frac{f(x)}{x^n} = \frac{f(x)}{x^m}\,x^{m-n} \to 0$ (as x^{m-n} is bounded near 0). Hence the first identity follows from item 1 of Proposition 8.1. The second and third are items 3 and 4 (with $h = x^k$ and $h = x^m$, respectively). The last one is item 5, since a polynomial $p(x) = a_0 + a_1 x + \cdots + a_d x^d$ is bounded on $(-1, 1)$ by $|a_0| + |a_1| + \cdots + |a_d|$. □

Example 8.2: As $x \to 0$ we have

$$\begin{aligned} \big(x + x^2 + o(x^2)\big)\big(1 - x + o(x)\big) &= x - x^2 + x^2 + \underbrace{x\,o(x)}_{o(x^2)} \underbrace{-x^3}_{o(x^2)} \\ &\quad + \underbrace{x^2\,o(x)}_{o(x^3)} + \underbrace{o(x^2)\big(1 - x + o(x)\big)}_{o(x^2)} \\ &= x + o(x^2). \end{aligned}$$

Here $-x^3 = o(x^2)$ since bigger powers are negligible near 0, $o(x^3)$ is also $o(x^2)$, and $1 - x + o(x)$ is bounded near 0 (it tends to 1). Note that the terms in x^2 cancel: the result is more precise than just $x + o(x)$.